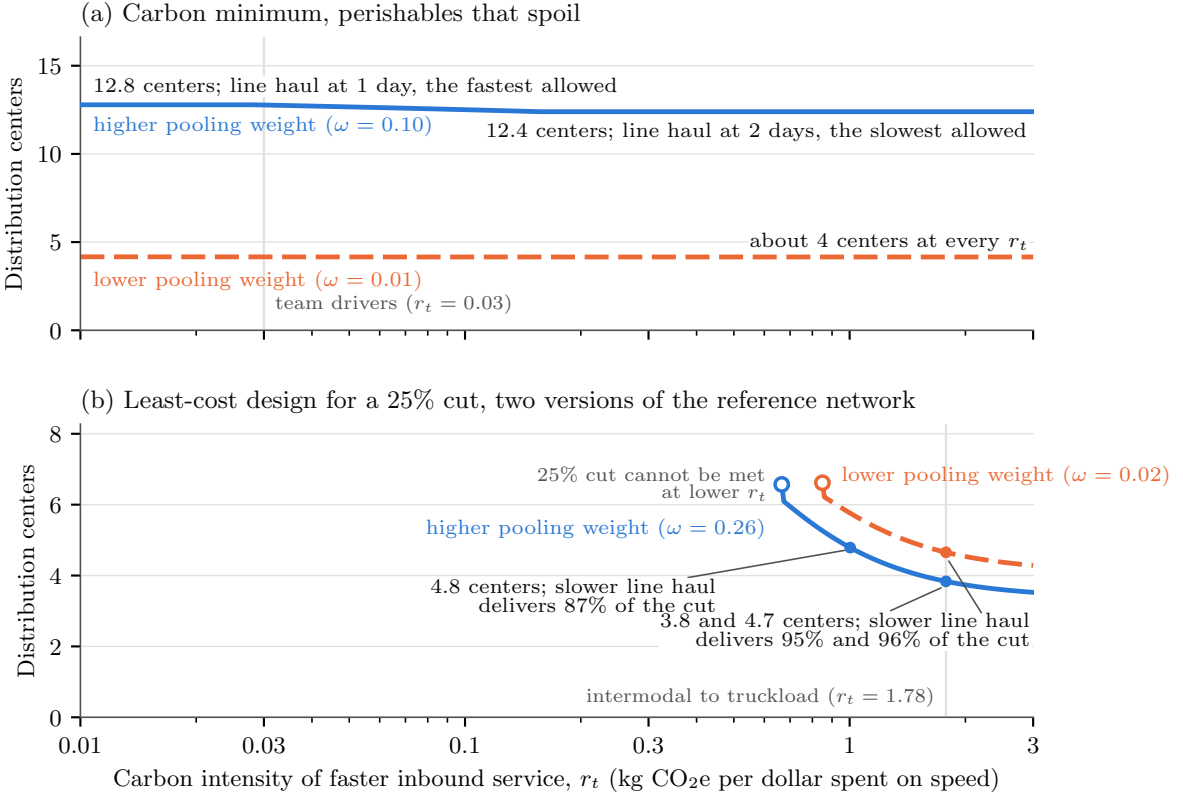

(a) Carbon minimum, perishables that spoil
Distribution centers
15
10
5
0
12.8 centers; line haul at 1 day, the fastest allowed
higher pooling weight ($\omega = 0.10$)
12.4 centers; line haul at 2 days, the slowest allowed
about 4 centers at every r_t
lower pooling weight ($\omega = 0.01$)
team drivers ($r_t = 0.03$)
(b) Least-cost design for a 25% cut, two versions of the reference network
Distribution centers
8
6
4
2
0
25% cut cannot be met at lower r_t
lower pooling weight ($\omega = 0.02$)
higher pooling weight ($\omega = 0.26$)
4.8 centers; slower line haul delivers 87% of the cut
3.8 and 4.7 centers; slower line haul delivers 95% and 96% of the cut
intermodal to truckload ($r_t = 1.78$)
0.01
0.03
0.1
0.3
1
3
Carbon intensity of faster inbound service, r_t (kg CO_2e per dollar spent on speed)